Final Project Theme

Data Explorer and Insight Generation Web Application

Objective

Develop a full-stack web application that allows users to:

1. Explore and search data.
2. Visualize information.
3. Generate statistics and insights.
4. Answer user questions through filtering and reporting.
5. Present data in an interactive and user-friendly manner.

Dataset Sources

Students may choose datasets from:

- Kaggle
- UCI Machine Learning Repository
- Data.gov
- WHO datasets
- World Bank datasets
- IMDb datasets
- FIFA datasets
- Netflix datasets
- COVID datasets
- Weather datasets
- Sales datasets
- Student performance datasets
- Spotify music datasets
- Airline datasets
- Car datasets
- Housing datasets

Example Project Ideas

1. Netflix Data Analysis System

Users can:

- Search movies
- Filter by genre
- View ratings
- Find top directors
- Analyze trends over years

Visualizations:

- Movies per year
- Genre distribution
- Rating distribution

Insights:

- Most common genre
- Highest-rated movies
- Growth of Netflix content

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2. Student Performance Dashboard

Users can:

- Search students
- Compare marks
- Filter by gender and subjects
- Analyze attendance

Visualizations:

- Average marks
- Pass/fail ratio
- Subject-wise performance

Insights:

- Best-performing subjects
- Factors affecting marks

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3. Car Price Analysis System

Users can:

- Search cars
- Filter by brand and year
- Compare prices

Visualizations:

- Price distribution
- Fuel type comparison

Insights:

- Which factors affect price?
- Average price by company

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4. FIFA Player Analytics

Users can:

- Search players
- Compare players
- Filter by age and nationality

Visualizations:

- Position distribution
- Age vs overall rating

Insights:

- Top countries
- Most valuable players

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5. Weather Analytics Portal

Users can:

- Search cities
- Filter dates

Visualizations:

- Temperature trends
- Rainfall charts

Insights:

- Hottest month
- Average annual rainfall

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6. Sales and Profit Dashboard

Users can:

- Filter products
- View regions
- Analyze sales

Visualizations:

- Monthly sales
- Category-wise profit

Insights:

- Highest profit products
- Best regions

Minimum Functional Requirements

User Management

- Registration
- Login
- Logout
- Profile page

Dataset Module

- Store dataset in database or CSV
- CRUD operations

Search and Filtering

Users should be able to:

- Search records
- Apply multiple filters
- Sort results

Dashboard

Should contain:

- Summary cards
- Charts
- Statistics

Examples:

- Count
- Average
- Maximum
- Minimum
- Top 10 records

Visualization Module

At least 5 charts:

1. Bar chart
2. Pie chart
3. Line chart
4. Histogram
5. Scatter plot

Libraries:

- Chart.js

- Plotly
- Matplotlib

Insight Generation Module

You should answer questions like:

- Which category has maximum value?
- What are the trends?
- Which region performs best?
- What is increasing or decreasing?
- What patterns can be observed?

The application should automatically generate these insights.

Example:

"Drama is the most common genre representing 32% of all movies."

"Sales in 2024 increased by 15% compared to 2023."

Reporting Module (any of the following)

Generate:

- HTML reports
- PDF reports
- CSV export

User Interface

Must include:

- Responsive design
- Navigation menu
- Search bar
- Pagination
- Tables
- Charts

ets.

Suggested Technologies

Backend

- Django

Frontend

- HTML
- CSS
- Bootstrap
- JavaScript

Database

- SQLite or MySQL

Data Processing

- Pandas
- NumPy

Visualization

- Chart.js
- Plotly

Deliverables

Students must submit:

1. Source code.
2. Dataset.
3. Screenshots.
4. User Manual.
5. Presentation slides.

How to submit

Place the project in your Google drive and submit the link at LMS.